# Automatic Noise Identification in Images Using Moments and Neural Network

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***Abstract*—Identifying noise from the original image is still a challenging research in image processing and is essential in order to counter the effects of unnecessary filtering process. Noise gets added to an image during image capture, transmission, or processing and degrades the performance of any image processing algorithms. Prior to de-noising step, the image should be tested for the identification of noise. Though Several approaches have been introduced in literature earlier for noise identification, each has its own assumption, advantages are not generic. This paper proposes a novel method based on statistical features with neural network classifier to identify the different types of noises such as Additive white Gaussian Noise, Salt & pepper Noise, Speckle Noise in the image without the human intervention. Extensive simulations on variety of images show that the proposed method effectively identifies the noise in a given image.**

***Keywords-Additive White Gaussian Noise, Salt & pepper Noise, Speckle Noise, Statistical features,Neural network.***

## I. INTRODUCTION

Noiseless images are intended for good interpretation of the data present in it. But noise gets added due to many factors including thermal effects, acquisition condition, sensor attenuation, A to D conversion, quantization etc. These noises in the images are unwanted information that disturbs the data present in it. So elimination of noise is one of the key research works to be done in computer vision and image processing which is fundamental to subsequent image processing technique such as edge detection, image segmentation, object recognition etc. Only a few researchers have worked on automatic noise identification [1],[2],[3] based on statistical parameters, [4] based on graphical methods, [5] based on gradient function methods in the images with the assumption that the given image is corrupted by noise. Thus identifying the presence of noise without human intervention lies prior to de-noising step. Usually neural network has been used for image classification and recognition. This paper proposes a neural network approach with third and fourth order moments as feature sets to identify the noise in an image and r is organized as follows section II describes the different types of noise model, section III describes the proposed methodology, section IV contains neural network training and classification, section V contains the results and discussion and section VI gives the conclusion.

## II. NOISE MODELS

Image noise is a random variation of brightness or color information in images produced by sensor circuitry of scanner or digital camera. It is an undesirable byproduct of image capture that adds spurious and extraneous information. Three types of common noise like Gaussian noise, Salt & pepper noise and Speckle noise have been taken for study.

Gaussian noise is statistical noise that has its probability density function equal to that of the normal distribution. The mathematical model is given is

$$g(i,j) = f(i,j) + n(i,j) \quad (1)$$

Where $1 \le i \le M$ and $1 \le j \le N$.
M, N is the size of the original image.

*Speckle noise is a Multiplicative noise*

Mathematical model for this noise is

$$g(i,j) = f(i,j)*n(i,j) \quad (2)$$

Where $1 \le i \le M$ and $1 \le j \le N$.

M, N is the size of the original image.

*Salt and pepper noise(impulsive noise)*

It is sometimes called as 'spike noise'. An image containing this type of noise will have dark pixels in brighter regions and bright pixels in darker regions. Mathematical model for this noise is

$$g(i,j) = \begin{cases} h_j & \text{with probability } h \\ f_j & \text{with probability } 1\text{-}h \end{cases} \quad (3)$$

Let '$f_j$' be the gray level of the original image 'f' at pixel location (i, j), '$h_j$' be the random numbers and 'h' is the noise ratio.

## III. PROPOSED METHOD

The given input image may or may not be corrupted by noise. The noise gets added to an image during many situations as

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discussed above in introduction. Image de-noising and enhancement is an important process in image processing for better interpretation of the data. Hence identification of noise prior to de-noising will reduce the effects of unnecessary filtering. The flow chart to identify the noise in images is shown in Figure 1.

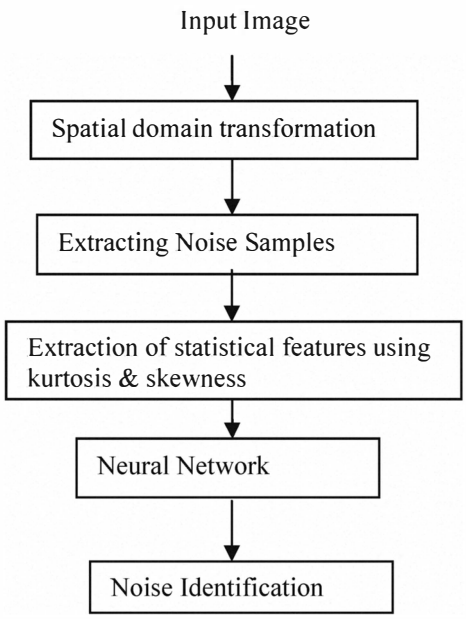


Figure 1: Flow chart of the proposed methodology

To identify the corrupted image, the image is converted into spatial domain. using wiener, median and lee filters [6].
The proposed method consists of three main steps to identify the noise type. Extract respective noise samples from the given noisy image. Estimate statistical features such as Kurtosis & Skewness. Well known spatial filters have been used to estimate noise samples from the given query image.

*A. Spatial Filter*

The primary objective of the image enhancement is to adjust the digital image so that the resultant image is more suitable than the original image for a specific application [5, 8, and 9]. There are many image enhancement techniques. They can be categorized into two general categories. The first category is based on the direct manipulation of pixels in an image, for instance: image negative, low pass filter (smoothing),and high pass filter (sharpening). Second category is based on the position manipulation of pixels of an image, for instance image scaling.spatial domain can be expressed as

$g(x, y) = T(f(x, y))$

Where, T is the transformation function, f (x, y) is the pixel value of input image, and g(x, y) is the pixel value of the processed image [6].The median, mean, high pass, low pass filtering techniques have been applied for denoising the different images [7].
We have to select three filters such as wiener filter, median filter and Lee filter are applied to the noisy image to get three different estimates of given image as follows:

$$\begin{aligned} g_{wiener}(i,j) &= f(I,j) * t_{wiener}(i,j) \\ g_{median}(i,j) &= f(I,j) * t_{median}(i,j) \\ g_{lee}(i,j) &= f(I,j) * t_{lee}(i,j) \end{aligned} \quad (4)$$

wiener filter is applied to identify the gaussian noise in an given image like median filter[7] for salt and pepper noise[7] and Lee filter for speckle noise. Gaussian and salt and pepper are additive noise but the speckle is the multiplicative noise. lee filters [11]are changed the multiplicative to additive noise. After apply the filters we have to extract the noise samples from the given image. Thus the noise samples obtained as follows

$$\begin{aligned} g_{wiener}(i,j) &= f(I,j) - g_{wiener}(i,j) \\ g_{median}(i,j) &= f(I,j) - g_{median}(i,j) \\ g_{lee}(i,j) &= f(I,j) - g_{lee}(i,j) \end{aligned} \quad (5)$$

*B. Extraction of statistical features*

The statistical features such as mean, standard deviation and moments are the most common to characterize a data set. Moments are calculated for all the input data sets . The mean and standard deviation provide information on the location, dispersion of the data.
Mean is the standardized first central moment of the probability distribution.

$$\mu = \frac{1}{M * N} \sum_{i=1}^{M} \sum_{j=1}^{N} S(i, j) \quad (6)$$

Where M, N-row and column of the image. S (i, j)-pixel value.
Standard deviation indicates the amount of variation or dispersion existing from the average mean.

$$\sigma = \sqrt{\frac{\sum_{i=1}^{M} \sum_{j=1}^{N} (S(i, j) - \mu)^2}{M * N}} \quad (7)$$

Where M, N-row and column of the image.
S(i, j)-pixel value

*C. Kurtosis & skewness*

Kurtosis is the standardized fourth central moment of probability distribution. It is a measure of peakedness of the distribution relative to Gaussian distribution.
The presence of speckle noise corrupts SAR images considerably and may conceal significant details on the images, leading to the loss of essential information. The first step toward despeckling is to identify the speckle noise in the SAR image. Kurtosis is estimated the presence of speckle noise in SAR image. Kurtosis and skewness is described as [8]

$$K = \frac{\sum_{i}^{j} S(i,j) - \mu^{(4)}}{\sigma^{(4)}} \quad (8)$$

$$S = \frac{\sum_{i}^{j} S(i,j) - \mu^{(3)}}{\sigma^{(3)}} \quad (9)$$

Kurtosis and skewness is calculated for the noise samples and used as the statistical feature in an neural network.These feature feed into the neural network classifier as the input.


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## IV. NEURAL NETWORK TRAINING AND CLASSIFICATION

Multilayer feed forward network is taken as the network architecture for the present application. The number of neurons in each layer and the number of hidden layers are decided on choosing the network. It is chosen by trial and error method based on the performance function until the specified goal is reached. Statistical moments kurtosis and skewness are the input variables to the input layer, having one hidden layer with nodes one greater than the nodes in input layer. The weights between the layers are initialized randomly [15] and the weights are adjusted by means of back propagation[16] on training the neural network

## V. RESULTS AND DISCUSSION

In this paper, DARPA Moving and Stationary Target[12-14] Acquisition and Recognition (MSTAR) SAR image database and real time photographic images is used[4].Noise samples have extracted from the input images as shown in figures 3 to 5 Features are extracted using skewness and kurtosis as shown in table1.The performance of the proposed algorithm is tested for 20 SAR images from MSTAR databases and 50 non SAR images compared with other classifier k-nearest neighbour (KNN) as shown in table2 and figure6 shows simulation results of neural network classifier. The maximum number of epochs is 3000.

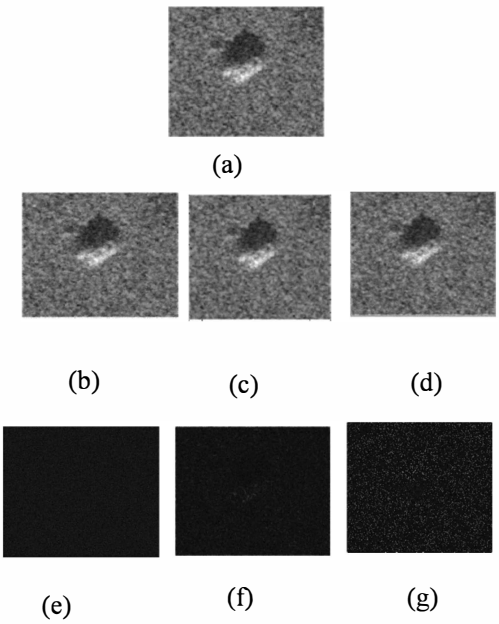


Figure 3.(a)Input SAR Image (BMP2) (b-d)applying wiener, median and Lee filter (e-g) Gaussian, Salt&pepper, speckle noise samples

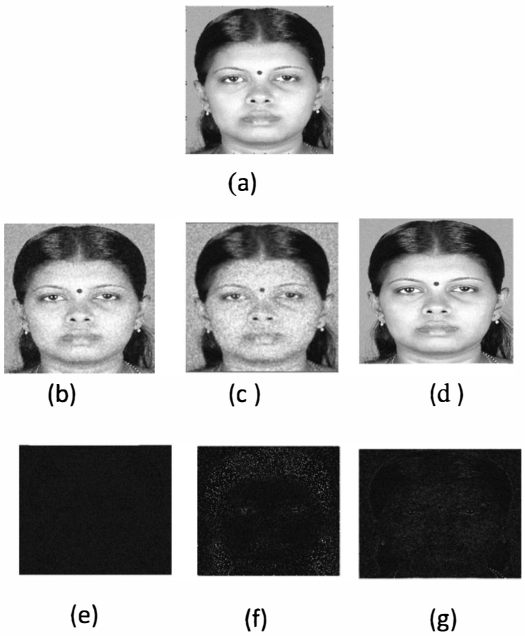


Figure 4.(a)Input photographic image-vasuki (b-d)applying wiener, median and Lee filter (e-g) Gaussian, Salt&pepper, speckle noise samples

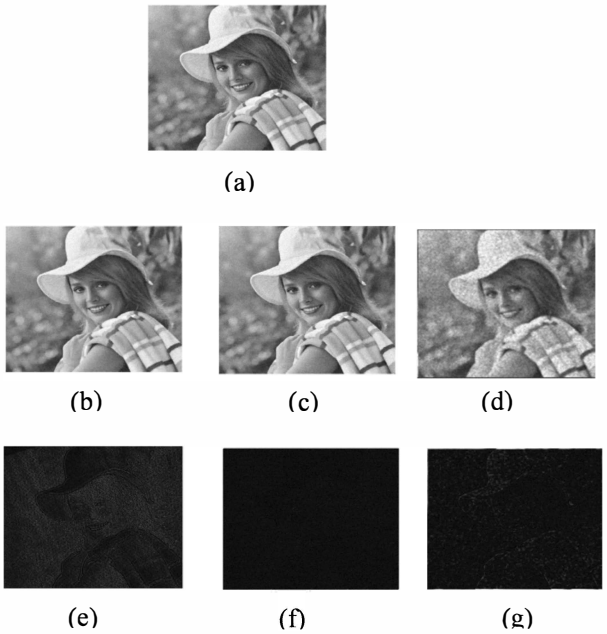


Figure 5.(a)Input photographic image (b-d)applying wiener, median and Lee filter (e-g) Gaussian, Salt&pepper, speckle noise samples

Table1.sample table of statistical features

| Input Image | Kurtosis Wiener | Kurtosis Median | Kurtosis Lee | Skewness Wiener | Skewness Median | Skewness Lee |
|---|---|---|---|---|---|---|
| | 4.6010 | 19.9331 | 26.2433 | 1.4973 | 4.1785 | 3.7212 |
| | 6.1991 | 1.7961 | 31.3557 | 5.1068 | 4.9808 | 1.5788 |
| | 6.4723 | 1.7867 | 19.2690 | 4.0726 | 4.7182 | 1.5189 |



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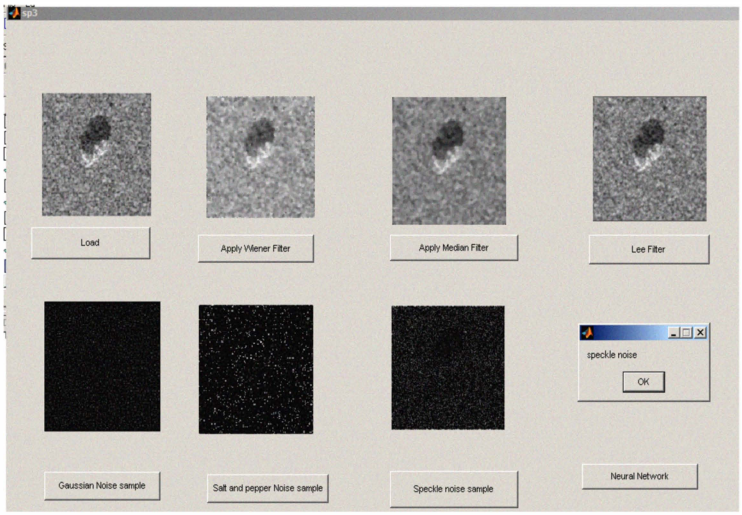
Table 2.Identification Accuracy

| Classifier | Images Tested ( η) | Gaussian Noise | | | Salt & Pepper | | | Speckle Noise | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Correct Identification ($\xi_c$) | Wrong Identification ($\varphi$) | Accuracy( $\xi$ %) | Correct Identification ( $\xi_c$ ) | Wrong Identification ($\varphi$) | Accuracy( $\xi$ %) | Correct Identification ( $\xi_c$ ) | Wrong Identification ($\varphi$) | Accuracy ( $\xi$ %) |
| K-NN | 70 | 63 | 07 | 90 | 62 | 08 | 88.57 | 61 | 05 | 87.14 |
| Neural Network | 70 | 69 | 01 | 98.57 | 65 | 05 | 92.85 | 69 | 01 | 98.57 |

Figure.6 simulation results of neural network classifier




## VI. CONCLUSION

An algorithm to identify the presence of noise in an image based on statistical moments and neural network is proposed. The three main types of noises such as Gaussian, salt and pepper, speckle are mainly considered and used for experimentation at various amount of noise. The proposed frame work is validated with accuracy rate (ξ) and wrong identification (φ) as shown in equation (10) and (11).

$$\xi = \frac{\xi_c}{\eta} \times 100 \ \% \qquad (10)$$

$$\varphi = 100 - \xi \ \% \qquad (11)$$

Where $\xi_c$ - no of images correctly identified, η – total of images tested. 98.57% accuracy of Gaussian and speckle noise shows the robustness of the proposed work. Accuracy of 92.85 for the salt and pepper noise. Extension of this work would include appropriate filtering action for the identified noise .

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